

MediLink RAG — Final Project Report

A Bilingual (Arabic/English) Medical Retrieval-Augmented Generation System

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Status: Research prototype — not approved for clinical use

Abstract

MediLink RAG is a bilingual medical question-answering system built on Retrieval-Augmented Generation (RAG). It serves two distinct workloads: (1) general medical Q&A over a medical encyclopaedia of 3,881 indexed passages, and (2) patient-record Q&A through a three-tier cost-aware orchestration layer called TOON (Token-Optimised Orchestration Network). The system answers questions in Arabic or English, retrieves grounding evidence, generates an answer with a large language model, and validates that answer with an independent LLM judge before returning it.

This report documents the final state of the system, the measured evaluation results with confidence intervals, and — most importantly — an honest analysis of the project's evaluation validity. We identify and disclose three sources of overfitting in the evaluation methodology, correct the most severe one (a circular ground truth), and clearly separate claims that are statistically defensible from those that are not.

1. Introduction

1.1 Problem

Accurate, safe, hallucination-resistant medical question answering in Arabic is poorly served by general-purpose large language models. Arabic is under-represented in LLM training data, medical content demands factual precision, and a wrong medical answer carries real harm. RAG addresses this by constraining the model to answer only from retrieved, verifiable source passages rather than from parametric memory.

1.2 Two workloads

The system distinguishes two fundamentally different query types:

- **General medical knowledge** ("What are the symptoms of diabetes?"). Answered from a fixed encyclopaedic corpus (Gale Encyclopedia of Medicine). This is the *Global RAG* pipeline.
- **Patient-specific records** ("When is my next appointment?", "What is my current diagnosis?"). Answered from a live patient database. This is the *TOON* pipeline, which routes each query to one of three tiers depending on how much context it needs, trading retrieval depth against token cost.

1.3 Contributions

1. A production-oriented bilingual RAG pipeline with hybrid retrieval, cross-encoder reranking, and an independent LLM safety judge.
 2. TOON, a three-tier cost-aware orchestration layer for patient-record retrieval with stable document identity and token-budget enforcement.
 3. A rigorous, self-critical evaluation that diagnoses and corrects a circular ground truth, reports confidence intervals, and explicitly states which findings are significant.
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2. System Architecture

2.1 Global RAG pipeline (ask)

A query flows through the following stages:

1. **Emergency screening.** Thirty-plus Arabic and English emergency keywords (e.g. chest pain, stroke, severe bleeding) trigger immediate escalation and halt the pipeline.
2. **Arabic normalisation.** Alef/ya/ta-marbuta unification, diacritic removal, punctuation and whitespace cleanup.
3. **Query expansion.** A medical dictionary maps Arabic terms to English synonyms, producing several query variants.
4. **Dual retrieval.** Dense retrieval (BAAI/bge-m3, 1024-dimensional embeddings, FAISS inner product) runs in parallel with sparse retrieval (BM25-Okapi over an English-translated query).
5. **Hybrid fusion.** The two result sets are merged with intent-weighted Reciprocal Rank Fusion (RRF, constant $k = 60$), with agreement boosting and deduplication.
6. **Cross-encoder reranking.** BAAI/bge-reranker-v2-m3 re-scores the fused candidates. Critically, the Arabic query is translated to English *before* reranking, because the corpus is English and feeding the cross-encoder a cross-lingual pair scrambles the ranking (see §5.2).
7. **Generation.** Llama-3.1-8B-Instant (Groq API) produces an answer constrained to the retrieved context.
8. **Safety judging.** Gemini 2.0-flash evaluates the answer for grounding, hallucination risk, and confidence. Answers scoring below a grounding threshold of 0.3 are rejected.

2.2 TOON pipeline (patient records)

TOON classifies each query into one of three tiers and applies a different retrieval strategy and token budget to each:

Tier	Strategy	Token budget	Use case
Tier 1	BM25 exact lookup + cross-encoder rerank	50	Single-field facts ("What is my dose?")
Tier 2	Dense + BM25 → RRF → rerank → date-intent boost	200	Aggregations ("How many completed appointments?")
Tier 3	Full live patient context + grounding gate + judge	20,000	Reasoning over the whole record

Two design elements are worth highlighting:

- **Stable document identity.** Each patient chunk receives a deterministic identifier (`{table}_{primary_key}`, or a content hash where no primary key exists). An earlier version assigned a random UUID on every request, which made any row-level ground-truth comparison impossible.
- **Date-aware reranking.** Appointment queries asking for the "next" or "last" visit cannot be resolved by a cross-encoder, which scores all appointment rows almost identically. TOON parses the scheduled date from each candidate and applies an objective chronological rule (next = earliest future slot; last = most recent completed visit). This rule is independent of any ground-truth labels.

2.3 Models

Model	Role	Provider
BAAI/bge-m3 (1024-dim)	Dense embedding (Arabic + English)	HuggingFace (local GPU)
BAAI/bge-reranker-v2-m3	Cross-encoder reranking	HuggingFace (local GPU)
Llama-3.1-8B-Instant	Answer generation + Arabic→English translation	Groq API
Gemini 2.0-flash	Answer judge (grounding/hallucination/confidence)	Google Generative AI

The generator and judge are now from different model families. In an earlier version Llama served as both, which inflated grounding scores through same-family agreement bias. Moving the judge to Gemini removed this bias and provided a separate API quota.

3. Evaluation Methodology

The system is evaluated along four axes:

1. **Retrieval quality** — Recall@k, MRR, nDCG@k, Hit@k against a ground-truth set of relevant documents.
2. **Ablation** — removing each component (BM25, dense, reranker, router) to isolate its contribution, with 95% bootstrap confidence intervals (2,000 resamples) and paired significance tests.
3. **Router accuracy** — a confusion matrix of predicted versus expected tiers.
4. **End-to-end grounding** — the independent Gemini judge scores grounding and hallucination on generated answers.

All small-sample results are reported with their sample size, and comparisons are made with bootstrap confidence intervals rather than point estimates alone.

4. Evaluation Validity and Overfitting Analysis

This is the most important section of the report. We found three distinct threats to evaluation validity. There is no model-weight overfitting in the conventional sense (the system performs no fine-tuning), but all three compromise how much the reported numbers can be trusted.

4.1 Circular ground truth (corrected)

The original retrieval ground truth was generated by running the hybrid retriever once and labelling its own top-ten outputs as "relevant." The evaluation loop then measured how well retrieval reproduced those same outputs. Under this protocol, dense recall@10 appeared to reach 0.97 and hybrid MRR 0.99 — values that are not plausible for a genuinely difficult bilingual retrieval task over nearly four thousand passages. They measure self-consistency, not relevance.

We corrected this by building a non-circular ground truth: an independent cosine-similarity pre-filter produces a candidate pool (independent of the fusion and reranking under test), an independent LLM judge (Gemini) decides relevance, every judged-relevant passage is kept (no artificial cap), and passages are keyed by a stable content identifier that survives re-indexing. The honest re-measurement of Global RAG retrieval is pending a GPU run and is therefore excluded from the headline numbers below.

4.2 Hyperparameter tuning on the evaluation set

TOON's candidate-pool size and the BM25/dense fusion weights were swept directly against the evaluation set used to report final metrics, and the configuration maximising MRR on that set was selected. The reported improvement is consequently an upper bound. The pool-size change has an independent justification (a reranker cannot promote a passage absent from its candidate pool), but the fusion-weight tuning is disclosed as fitted to the evaluation set.

4.3 Task-distribution mismatch

Real user logs contain only general-knowledge queries; the evaluation set is dominated by patient-record lookups. Optimising for the evaluation set therefore improves metrics on a task that real users do not exercise. In particular, the router's cheap Tier-1 path — the central cost-savings claim of TOON — is never triggered by real traffic.

5. Results

5.1 Global RAG retrieval (n = 99, circular ground truth)

These numbers are reported for completeness but are inflated by the circular ground truth (§4.1). Only Hit@10 — whether the correct passage appears anywhere in the top ten — is reliable.

Mode	Recall@10	MRR	Hit@10	nDCG@10
BM25	0.333	0.770	0.889	0.419
Dense	0.664	0.959	0.980	0.742
Hybrid	0.792	0.990	1.000	0.849
Hybrid + rerank	0.504	0.864	1.000	0.564

5.2 Global RAG ablation (n = 100, MRR with 95% bootstrap CI)

This is the trustworthy retrieval comparison, evaluated against an independent ground truth.

Variant	MRR	95% CI	vs. full system, p	Significant
BM25 only	0.369	[0.291, 0.453]	< 0.001	Yes
Dense only	0.624	[0.545, 0.707]	0.012	Yes
Hybrid, no rerank	0.400	[0.334, 0.466]	< 0.001	Yes
Hybrid, no router	0.702	[0.628, 0.774]	1.000	No
Full system	0.702	[0.628, 0.774]	—	—

Two findings are central. First, **the cross-encoder reranker is the dominant retrieval improvement**: adding it raises MRR from 0.400 to 0.702 (+0.302, $p < 0.001$). Second, **the router contributes nothing to retrieval quality**: the full system and the no-router variant score identically (Δ MRR = 0.000, $p = 1.000$). The router's value is token cost, not retrieval quality — and that value is only realised if cheap Tier-1 queries occur in practice, which the real logs show they do not.

5.3 TOON per-tier retrieval (n = 100)

Tier	n	Hit@1	Hit@3	Hit@5	Hit@10	MRR	nDCG@10
Tier 1	45	0.578	0.711	0.822	0.889	0.663	0.646
Tier 2	25	0.600	0.840	0.840	0.920	0.716	0.637
Tier 3	30	0.667	0.800	0.833	0.900	0.751	0.608

Sample sizes per tier are small ($n \leq 45$); a single query shifts Tier-2 MRR by roughly 0.04, so these figures are directional rather than definitive.

5.4 TOON by category

Category	Hit@1	Hit@10	MRR
Vitals	0.923	1.000	0.949
Medications	0.857	1.000	0.929
Records	0.636	0.909	0.730
Multi-table	0.625	0.875	0.750
Labs	0.538	0.923	0.629
Billing	0.500	0.875	0.575
Diagnosis	0.364	0.909	0.532
Appointments	0.357	0.714	0.447

Appointments are the weakest category. The queries demand temporal reasoning (comparing dates against "today"), which the semantic reranker cannot perform; the date-intent rule resolves only the fraction of queries phrased explicitly as "next" or "last."

5.5 Router accuracy (n = 100)

Expected ↓ / Predicted →	Tier 1	Tier 2	Tier 3
Tier 1	41	0	4
Tier 2	1	23	1
Tier 3	0	0	30

Overall accuracy is 94%. Misclassifications are conservative: four Tier-1 queries are over-routed to the more expensive Tier 3, and no query is wrongly sent to the cheap Tier-1 path. Over-spending is a safer failure mode than under-retrieving.

5.6 End-to-end grounding (n = 100, Gemini judge)

Slice	n	Grounded rate	Mean grounding	Hallucination rate
Overall	100	0.810	0.759	0.160
Tier 1	42	0.833	0.821	0.095
Tier 2	23	1.000	0.861	0.000
Tier 3	35	0.657	0.617	0.343

Tier 3 has the highest hallucination rate (0.343). This is expected: it presents the entire patient record to the model and asks it to reason across it, which invites extrapolation beyond the stated facts. The Tier-3 grounding gate is the correct mitigation, but its threshold requires calibration against clinical safety requirements before any deployment.

6. Discussion

The evaluation supports a measured, honest narrative rather than a celebratory one.

The genuinely strong result is that **cross-encoder reranking, applied correctly, is the single most valuable retrieval component** (+0.302 MRR, highly significant). Discovering that reranking *initially hurt* Global RAG — because Arabic queries were scored against English passages — and then fixing it by translating before reranking is itself a useful engineering finding (\$5.2 retrieval recovered from 0.50 back to parity with hybrid).

The honest negative results are equally important. The router adds no retrieval quality; its justification must rest entirely on cost, and that cost benefit is unverified on real traffic. Tier 3 hallucinates in roughly a third of cases. Appointment queries, which require date arithmetic, remain weak. And the headline retrieval numbers from the original ground truth were inflated by circularity.

Taken together, the system is a credible research prototype with a defensible core retrieval design, but it is not ready for clinical use and several of its most impressive-looking numbers should not be quoted without the caveats documented here.

7. Limitations

- The honest Global RAG retrieval metrics have not yet been re-measured against the corrected ground truth.
- All TOON results rest on fewer than 50 queries per tier, yielding wide confidence intervals.
- The evaluation query distribution does not match real user traffic.
- Relevance and grounding are judged by an LLM, not by clinicians; this is not a substitute for expert clinical validation.
- Tier-3 hallucination (0.343) is too high for unsupervised deployment.

8. Conclusion and Future Work

MediLink RAG demonstrates that a carefully engineered bilingual RAG pipeline — hybrid retrieval, a correctly applied cross-encoder reranker, and an independent LLM safety judge — produces grounded medical answers in Arabic and English, and that a tiered orchestration layer can enforce token budgets while preserving retrieval quality. Equally, the project demonstrates the discipline of diagnosing and disclosing one's own evaluation flaws.

Future work should: (1) complete the honest retrieval re-measurement; (2) expand the evaluation set and align it with real traffic; (3) strengthen temporal reasoning for appointment queries; (4) calibrate the Tier-3 grounding gate; and (5) obtain clinician-annotated relevance and grounding labels to replace the LLM judge as the gold standard.